

PlantPal: Tutoring Pest and Disease Knowledge via Large Language Models

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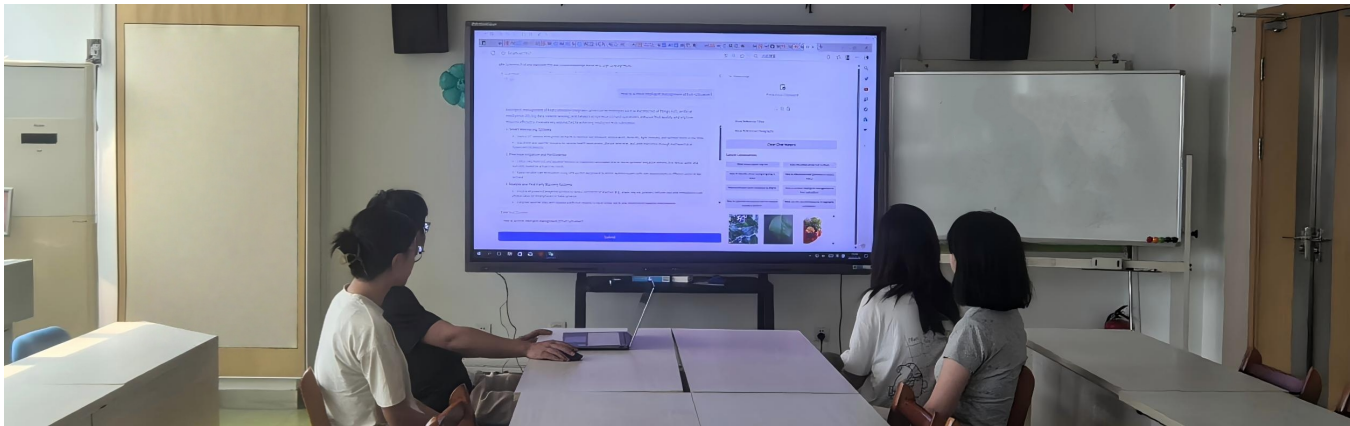


Figure 1: A full tutorial session engaging with PlantPal activities.

Abstract

Knowledge of pests and diseases in agriculture and forestry has been conveyed through static texts and two-dimensional images. However, traditional static resources lack interactivity and contextual reasoning support to guide students better master and remember these knowledge. To address these challenges, we present PlantPal, an interactive question-answering system powered by large language models (LLMs) that enables dynamic reasoning and multimodal interpretation of such knowledge. This system illustrates the potential of LLM-based approaches to enhance agricultural students' cognitive engagement and support the translation of theoretical understanding into practical application. By developing a knowledge representation framework that balances domain specificity with accessibility, the system effectively communicates complex concepts, while remaining aligned with the foundational cognitive level of undergraduate students in agricultural education. Results from extensive user studies, involving 50 undergraduate agricultural students, demonstrate statistically significant improvement in students' comprehension and practical reasoning, with

participants reporting increased motivation and engagement compared to traditional text-based learning resources. This work contributes to the ongoing transformation of agricultural pedagogy by introducing a new paradigm of immersive cognitive intelligence.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction.**

Keywords

Large Language Model, Agricultural and Forestry Education, Plant Pests and Diseases, Deep Learning

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1 Introduction

The rapid advancement of generative large language models (LLMs) is reshaping educational paradigms through dynamic content generation and cognitive interaction reconstruction, thereby significantly deepening their applications in education and training. Many studies have already certified that task training systems driven by LLMs have achieved cross-domain knowledge transfer. For instance, in the field of climate prediction, GraphCast [Lam et al. 2023] is capable of generating global high-resolution meteorological evolution maps, with accuracy surpassing traditional numerical models. In the field of new material research and development, AlphaFold 3

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can generate protein-ligand binding conformations, accelerating the process of biomaterial design [Abramson et al. 2024]. In the field of legal intelligence, Lawyer GPT [Yao et al. 2024] can perform well in both subjective and objective legal domains, showing a strong ability to handle legal issues. In computer graphics education, GPT-4o is capable of solving university-level geometry and rendering questions through multimodal reasoning, indicating the growing reliability of LLMs in spatial and visual domains [Feng et al. 2024].

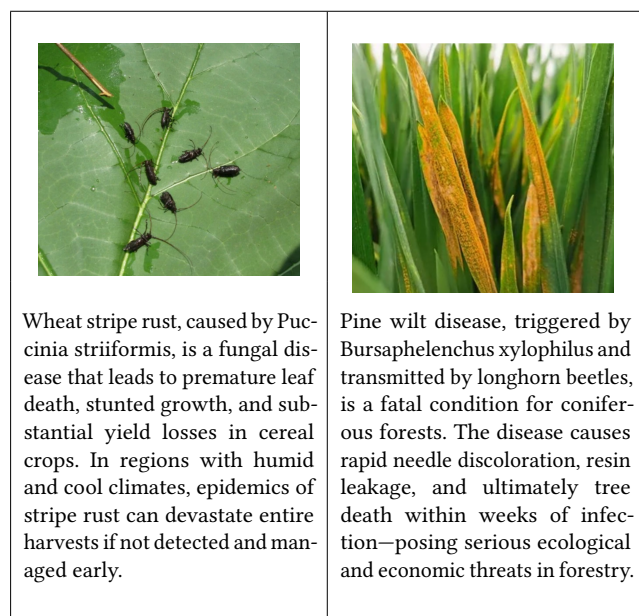


Figure 2: Two example images and their corresponding detriments.

Nonetheless, the potential of large language models (LLMs) to deeply interpret domain-specific knowledge in agriculture and forestry remains underexplored. The potential capacity of LLMs to address complex domain-specific knowledge challenges is increasingly regarded as a breakthrough for interdisciplinary educational innovation, with agricultural and forestry pest and disease knowledge systems representing a particularly promising area of application [Tzachor et al. 2023]. This potential is particularly salient in the context of professional education for agricultural and forestry majors at universities. Examples (See Figure 2) underscore the complexity and urgency of pest and disease management, particularly for students and practitioners who must integrate visual symptom recognition, biological knowledge, and intervention strategies. However, traditional static resources, such as textbooks or isolated image databases, lack interactivity and contextual reasoning support. Nevertheless, the knowledge system related to agricultural and forestry pests and diseases is characterized by its large volume and a high degree of content redundancy, which poses considerable challenges for student memorization and understanding. Furthermore, textbook update cycles typically exceed five years, making it difficult to incorporate newly emerging diseases in a timely manner. This contributes to knowledge stagnation, one of the major

limitations of traditional teaching methods in this field. In contrast, generative large language models (LLMs) offer promising solutions for bridging the aforementioned knowledge gaps by enabling real-time knowledge updates and clarifying commonly misunderstood or overlapping concepts, particularly in educational settings.

The application adaptability of generative large language models (LLMs) in the domain of pest and disease management stems from their multimodal integration capabilities and strengths in complex system modeling. These models can synthesize satellite remote sensing data, pathological imagery, and field sensor inputs to enable dynamic, cross-scale inference of agricultural disasters. Notably, the domain's highly structured knowledge system aligns well with the cognitive architecture of LLMs. However, this field has long been constrained by two major challenges. At the data level, approximately 80% of critical agricultural expertise exists in unstructured formats, dispersed across field logs and local archives. At the technical level, conventional models struggle to jointly process two fundamentally heterogeneous data types: hyperspectral imagery and natural language descriptions. The release of Alibaba Cloud's Qwen [Bai et al. 2023] series of LLMs in late 2023 presents a promising technical breakthrough to address these longstanding limitations.

It is important to note that when large language models (LLMs) are applied directly to the domain of agricultural and forestry pest and disease management without domain-specific fine-tuning, they often exhibit substantial deficiencies in both knowledge adaptation and semantic understanding. The inherent priors of general-purpose foundation models frequently fail to capture the specialized terminology used in this field, which can lead to distorted interpretations of pathological mechanisms. Empirical studies have reported that the hallucination problem has become a major technical challenge in domain-specific question-answering systems, with its incidence rate being non-negligible [Huang et al. 2025]. In addition, these models commonly lack structured reasoning when responding to domain-specific instructions, and they may confuse the agronomic priorities involved in prevention and control strategies. Prompt-tuning [Lester et al. 2021], through the injection of structured domain knowledge, offers a viable solution by reconstructing the model's cognitive framework. This process embeds domain-specific terminology and reasoning patterns into the model's parameter space, thereby enabling decision-making pathways that are better aligned with the requirements of the agricultural and forestry domain.

We present the PDMA-YOLO and Prompt-Tuning-driven Integrated System for Pest and Disease Identification and Question Answering—an application based on large language models (LLMs), designed to support knowledge instruction related to pest and disease management in the agricultural and forestry domains, enabling learners to engage with pest and disease information in a dynamic and multimodal fashion.

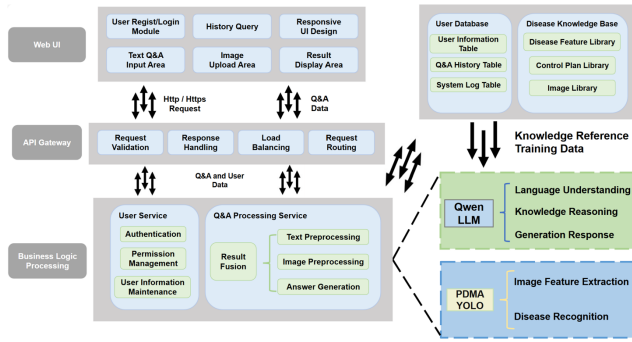
2 Methods

The central objective of this study is to harness the robust logical reasoning and generalization capabilities of large language models (LLMs) to analyze and interpret the complex feature information present in images of fruits and vegetables, with the goal of achieving

Table 1: Dataset Category Details

Plant Type	Number of Disease Categories	Plant Type	Number of Disease Categories
Apple	4	Peach	2
Cassava	5	Potato	3
Cherry	2	Rice	4
Corn	4	Squash	1
Grape	4	Strawberry	2
Orange	1	Tomato	6
Pepper	2	Total	40

more stable, accurate, and comprehensive plant disease diagnoses. The proposed approach integrates the Qwen large language model with PDMA-YOLO to extract image features from agricultural edge devices and convert them into textual representations for real-time diagnosis of plant diseases and pests. The overall network architecture is depicted in the accompanying figure (See Figure 3).

**Figure 3: The architecture of PlantPal.**

In this framework, agricultural images are first processed by a pre-trained YOLO network, which is employed to detect and identify visual features indicative of plant diseases and pests. The output from YOLO is subsequently transformed into textual descriptions and input into the LLM. Leveraging its advanced reasoning capabilities, the LLM synthesizes information from all YOLO outputs and relevant agricultural sensor data, ultimately generating diagnostic conclusions and corresponding treatment recommendations.

2.1 Dataset

This study constructs the Plant Disease Dataset, which encompasses 13 crop categories and 40 disease categories. Major categories are presented in the above table (See Table 1).

This dataset comprises 29,813 images and 52,638 annotated instances in total. Prior to model train, the dataset underwent stratified partitioning into training, validation, and testing subsets at a 7:2:1 ratio. This partitioning strategy employed class-aware sampling to maintain consistent distribution across partitions, ensuring robust evaluation of model generalization capabilities.

2.2 An Improved PDMA-YOLO

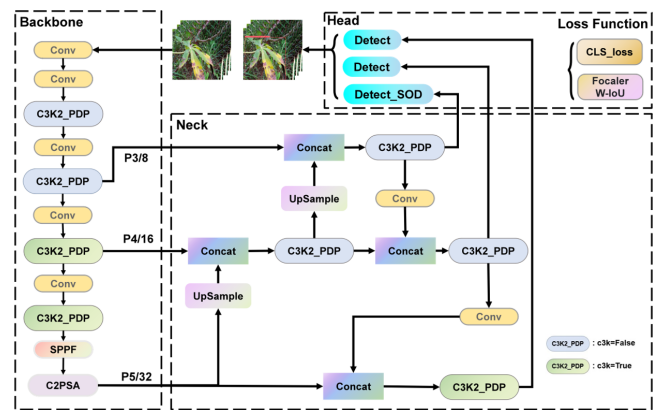
In early studies, researchers mostly used manual feature extraction techniques combined with traditional machine learning methods to achieve disease detection [Bekkerman and Tabrikian 2006; Nasrabadi 2014]. V. Singh et al. [Singh and Misra 2017] achieved 93% plant disease detection accuracy through image feature segmentation techniques and genetic algorithms. However, manual feature design is time-consuming and has poor generalizability, struggling to adapt to the diversity of disease symptoms and complex field environments, while also leading to a higher rate of missed detections for multi-scale disease targets.

The development of deep learning techniques has driven a transformation in plant disease detection. P. Johri et al. [Johri et al. 2022] developed the "FarmEasy" platform based on deep learning techniques, achieving accurate identification of leaf diseases. Fenghua Wang et al. [Wang et al. 2023] proposed YOLOv7-PD, which achieved 90.3% mAP in real-time fruit detection. Zeng Ti et al. [Zeng et al. 2023] achieved 93% mAP in tomato disease detection by improving YOLOv5. Among them, the YOLO series has demonstrated significant advantages in agricultural disease detection due to its efficient end-to-end detection capability.

The proposed YOLOv11 model [Khanam and Hussain 2024] is characterized by high inference speed, a lightweight architecture, and multi-class and multi-scale detection capabilities, making its framework an ideal foundation for targeted optimization in complex agricultural scenarios—thus was selected as the baseline for this study.

However, existing algorithms still face dual challenges of feature extraction accuracy and real-time detection efficiency in complex agricultural environments, making optimization and improvement of YOLOv11 crucial to fully unleashing its potential.

We present PDMA-YOLO (see Figure 4), designed to accurately and efficiently detect plant diseases in agricultural production. Built upon the traditional YOLOv11 model, the architecture consists of three main components: Backbone, Neck, and Head.

**Figure 4: The architecture of the PDMA-YOLO model.**

In the Backbone and Neck sections, the proposed PDP Bottleneck module is integrated with the original C3K2 module to enhance the model's feature extraction and integration capabilities. For the Head

component, to balance performance and accuracy, this paper introduces the MLLA attention mechanism to improve the small-object detection head. By combining the merits of Mamba architecture with linear attention, this mechanism enhances the model's sensitivity to subtle signs of disease or pest infestation, which are often challenging to detect in complex orchard or farm environments. In terms of the loss function, a novel Focaler-W-IoU loss is proposed by combining the advantages of Focaler-IoU and W-IoU. This loss function facilitates faster and more accurate target localization by emphasizing hard-to-detect samples and incorporating spatial information in bounding box regression.

2.2.1 PDP Bottleneck module: Balancing Feature Extraction and Computational Performance. PDP Bottleneck module is composed of depthwise convolution and pointwise convolution in a "Pointwise-Depthwise-Pointwise" bottleneck structure (See Figure 5). Specifically, the first pointwise convolution is employed to adjust the number of channels, reducing computational complexity while enabling inter-channel information interaction. This step compresses or expands the channel dimension, laying a foundation for efficient feature processing.

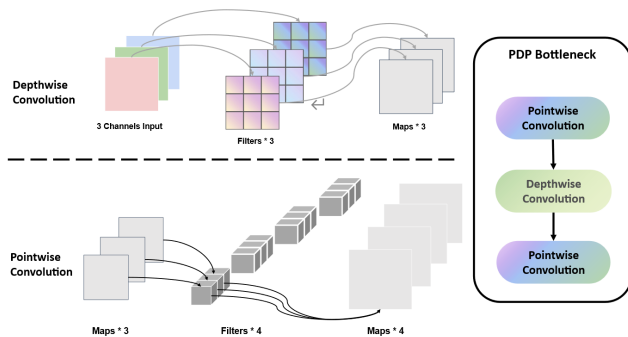


Figure 5: The architecture of the PDP Block.

Subsequently, the depthwise convolution operates on each channel independently, focusing on capturing spatial-domain features. Unlike traditional convolution that processes all channels together, depthwise convolution delves into each channel's spatial details, significantly enhancing the model's ability to extract spatial-related characteristics of fruit and vegetable pests and diseases.

Finally, another pointwise convolution is utilized to further adjust the channel dimension and integrate features across channels. This step ensures that the spatial features extracted by depthwise convolution are effectively combined with channel-wise information, achieving a comprehensive fusion of multi-dimensional features.

Compared with the original C3K2 module, the PDP Bottleneck module introduces this efficient bottleneck structure. It not only reduces the computational load by decomposing the convolution operations but also strengthens the feature extraction capability. In the Backbone and Neck of the model, this module allows for more refined feature extraction from input data and better integration of multi-scale or multi-aspect features. For example, in the Backbone, it can capture fundamental plant disease-related features from low-level representations, while in the Neck, it aids in fusing

features from different layers, ensuring that subsequent detection heads receive rich and discriminative feature representations. This structure optimizes the balance between computational efficiency and feature-processing performance, making the model more adept at handling complex detection tasks.

2.2.2 Mamba-Like Linear Attention (MLLA) for small object detection. The MLLA (Multi-Level Linear Attention) mechanism, proposed by D. Han et al. [Han et al. 2024], innovatively integrates the forget gate and linear attention, which can effectively enhance the performance of algorithms for detecting plant diseases.

MLLA achieves feature optimization through a dual mechanism (See Figure 6). Firstly, before being input into this module, the dynamic forget gating unit adopts a parametric selection strategy. It non-linearly modulates the feature channels using a learnable weight matrix, significantly enhancing the ability to identify key plant disease features while suppressing noise interference in non-significant regions. Second, the linear attention mechanism based on the kernel approximation method reduces the computational complexity from $O(N^2)$ in traditional Transformers to $O(N)$, enabling efficient modeling of the correlation relationships in the feature space. This combination enables high-precision detection in plant disease detection tasks and can accurately capture subtle pests and diseases features.

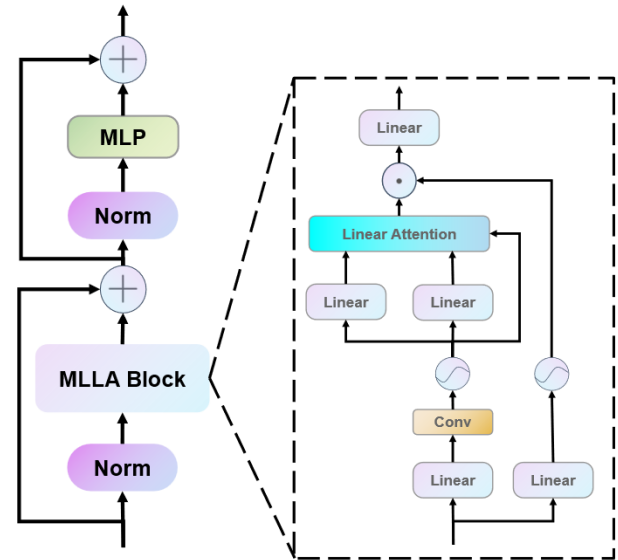


Figure 6: The architecture of the MLLA Block.

Considering that in actual plant disease detection, the disease blocks of plant disease instances are usually small, to avoid unnecessary computational overhead, we only added the MLLA attention to the small-object detection head of the YOLO model. This design allows the model to have superior computational performance while effectively addressing the challenges where both precise localization and rapid processing are crucial in plant disease detection. By integrating MLLA, a critical balance between accuracy and efficiency is achieved.

2.2.3 Focaler-W-IoU Loss Function. To optimize the model's positioning accuracy, assist the model in converging to the best performance more quickly, and address the class imbalance issue in practical scenarios, we introduced the Focaler-W-IoU loss function. This loss function consists of the Focaler [Lin et al. 2017] mechanism, which draws inspiration from the Focal Loss concept, and W-IoU [Tong et al. 2023].

When calculating the Intersection over Union (IoU), W-IoU incorporates the width information of the bounding boxes, comprehensively considering their geometric shapes and relative positional relationships. It can more accurately measure the distance and similarity between the predicted and ground-truth bounding boxes, guiding the model to generate more precise bounding box predictions and improving the positioning accuracy. Meanwhile, its ability to adaptively adjust the penalty strength helps the model converge to the optimal solution more rapidly, enhancing the accuracy and stability of positioning. The calculation formula is as follows:

$$L_{IoU} = 1 - IoU = 1 - \frac{W_i H_i}{S_u} \quad (1)$$

$$R_{IoU} = \exp \left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{((w_g)^2 + (h_g)^2)^*} \right) \quad (2)$$

$$L_{W-IoU} = \left(\frac{\mathcal{L}_{IoU}^*}{\mathcal{L}_{IoU}} \right)^Y \exp \left(\frac{(x - x_{gt})^2 + (y - y_{gt})^2}{(W_g^2 + H_g^2)^*} \right) \left(1 - \frac{W_i H_i}{S_u} \right) \quad (3)$$

L_{IoU} indicates the loss value of the intersection ratio. W_i and H_i represent the width and height of the area. They are used to calculate the cross-sectional area S_u . R_{IoU} is a regression factor related to the deviation of the center position of the bounding box. x and y represent the center point of the predicted bounding box, x_{gt} and y_{gt} represent the center point of the target bounding box. W_g and H_g represent the width and height of the minimum bounding rectangle of the target box, which is also the actual size of the target.

In actual agricultural production, the occurrence frequencies of different plant disease categories vary significantly. The resulting sample imbalance problem causes the model to more easily learn the features of common plant diseases during training while under-learning the features of rare plant diseases. The Focaler mechanism can reconstruct the W-IoU loss using a linear interval mapping method and then assign different weights to samples with different IoU values (the calculation process is as follows). It reduces the weights of easily-classifiable samples and increases the weights of difficult-to-classify samples, enabling the model to pay more attention to such challenging samples.

$$IoU^{Focaler} = \begin{cases} 0, & IoU < d \\ \frac{IoU - d}{u - d}, & d \ll IoU \ll u \\ 1, & IoU > u \end{cases} \quad (4)$$

$$L_{Focaler-W-IoU} = L_{W-IoU} + IoU - IoU^{Focaler} \quad (5)$$

2.3 The Design of Language Model

2.3.1 Selection of Qwen as the Base Model. This study adopts Qwen2.5-72B-Instruct as the foundational language model, based on

several key considerations. First, the model benefits from an extensive knowledge base, having been trained on a dataset comprising over 18 trillion tokens. Second, the 72B variant of Qwen2.5 has demonstrated superior performance, surpassing LLaMA3.1-405B in multiple benchmarks, establishing itself as one of the most powerful open-source large language models globally. Finally, Qwen2.5 supports over 29 languages, including Chinese, which makes it highly suitable for multilingual agricultural applications.

2.3.2 RAG-Based Optimization to Address Hallucination and Enhance Domain Knowledge Understanding. To address the challenges of hallucinations and limited access to up-to-date or domain-specific knowledge commonly encountered in large language models, this study introduces an optimized framework based on Retrieval-Augmented Generation (RAG). This approach significantly enhances the model's accuracy and reliability in understanding and generating domain-specific content.

A specialized knowledge base focused on fruit and vegetable diseases and pests was constructed and indexed using the efficient LlamaIndex framework. The knowledge base is stored in a vector database (VectorDB), which supports dense embedding-based semantic similarity search. During inference, relevant authoritative knowledge fragments, such as symptom descriptions, pathogen or pest details, control methods, and the latest research findings, are retrieved in real-time based on user queries. These contextual fragments are dynamically injected into the input of language model, guiding it to generate responses strictly grounded in verified domain knowledge.

Experimental results confirm that this method effectively reduces hallucinations and significantly improves the factual accuracy and timeliness of outputs related to pest diagnosis and treatment recommendation tasks. Furthermore, the method preserves the original strengths of the language model in understanding and reasoning, offering a technically sound and practically valuable solution for building reliable and knowledge-traceable intelligent Q&A systems in agriculture.

2.3.3 Efficient Fine-Tuning via Prompt-Tuning. This study employs Prompt-Tuning as a parameter-efficient fine-tuning strategy to adapt the large-scale pre-trained model to the specialized task of fruit and vegetable pest and disease diagnosis. Given the substantial size of the Qwen2.5-72B-Instruct model, traditional full-parameter fine-tuning would be computationally prohibitive. Therefore, Prompt-Tuning is used to insert a small number of trainable soft prompts (also known as prefixes) at the model's input layer, while freezing the remaining model parameters.

The model is fine-tuned on a curated dataset specific to fruit and vegetable diseases and pests, enabling it to internalize domain-specific knowledge and reasoning patterns. Experimental evaluations demonstrate that this approach significantly enhances performance on tasks such as disease identification, causal analysis, and treatment recommendation, while greatly reducing computational cost and storage requirements. This enables more accessible and high-quality intelligent support for students and technical personnel in agriculture.

2.4 The Integration of PDMA-YOLO and Language Model

To address the issue of excessive memory consumption when integrating large-scale models with agricultural systems, this study proposes a hybrid image-text fusion method based on Qwen's capabilities. The fusion of visual and textual modalities is achieved through a transformation-based approach.

Inference tasks requiring complex reasoning over visual information are offloaded to the cloud. Image features are first extracted and preliminarily mapped on edge devices, then converted into compact textual descriptions. This allows for the transmission of concise text segments rather than high-resolution images, thereby optimizing communication efficiency for real-time diagnosis.

To implement this, PDMA-YOLO is employed to extract visual features, which are then converted into text. These text fragments are passed into language model with task-specific prompts, enabling the model to summarize and organize potentially redundant information through its reasoning capabilities. The processed text is subsequently input into the language model, along with agronomic prompts, to perform diagnostic reasoning and provide interactive, immediate feedback.

This system significantly reduces data transmission overhead and enhances real-time performance, offering a practical and scalable solution for intelligent agricultural diagnostics.

3 Results

3.1 Evaluation Metrics

This study evaluates model performance using three principal metrics: Precision, Recall, and mean Average Precision (mAP). Use FPS and model size to reflect model's degree of speed and lightweight respectively.

The specific calculation formula is as follows:

$$\text{precision} = \frac{1}{h} \sum_{i=1}^h \frac{TP_i}{FP_i + TP_i} \quad (6)$$

$$\text{recall} = \frac{1}{h} \sum_{i=1}^h \frac{TP_i}{FN_i + TP_i} \quad (7)$$

$$\text{mAP} = \frac{1}{h} \sum_{i=1}^h \int_0^1 P_i(R_i) dR_i \quad (8)$$

$$\text{FPS} = \frac{1}{t_{\text{pre}} + t_{\text{infer}} + t_{\text{post}}} \quad (9)$$

Where h represents the number of behavior categories, and i represents specific behavior categories. TP_i represents the behavior of the model correctly classified into category i while FP_i represents the behavior of the model incorrectly classified into category i . FN_i indicates missing classification, indicating that the actual behavior belongs to category i , but the model incorrectly identifies it as other categories. R_i represents recall corresponding to category i . Preprocessing time, inference time, and postprocessing time are denoted by t_{pre} , t_{infer} and t_{post} respectively.

3.2 Comparative Experiments between Different Models

To validate the PDMA-YOLO model's effectiveness, comparative experiments were conducted on seven architectures—including two-stage (Faster-RCNN [Girshick 2015]), single-stage (YOLO series [Jocher et al. 2021; Varghese and M. 2024; Wang et al. 2024] and CenterNet [Duan et al. 2019]), lightweight (EfficientNet [Tan et al. 2020]), RT-DETR [Zhao et al. 2024] and baseline using a unified dataset and experimental protocols (see Table 2).

Table 2: Comparison of Various Models Across Diverse Behavioral Detection Results

Models	P	R	mAP	FPS	Size
YOLOv11	88.3	89.5	87.2	117	2.61MB
Faster-RCNN	47.9	81.3	77.1	46	9.76MB
EfficientNet	85.4	74.4	83.9	63	2.76MB
CenterNet	82.8	60.2	86.8	89	6.78MB
RT-DETR	90.8	87.7	92.2	52	32.4MB
YOLOv5	89.1	87.7	93.2	118	2.64MB
YOLOv8	88.9	92.3	94.5	114	2.67MB
YOLOv10	89.7	88.4	92.7	115	2.63MB
PDMA-YOLO	92.5	93.6	96.3	137	2.13MB

Among various models demonstrated in Table 2, the PDMA-YOLO model demonstrates outstanding performance. It achieves the highest precision ($P = 92.5\%$), recall ($R = 93.6\%$), and $\text{mAP} = 96.3\%$, indicating superior detection accuracy and comprehensive performance. Meanwhile, it maintains a high FPS of 137 and the smallest model size (2.13MB), achieving an excellent balance between speed and lightweight design.

In contrast, other models have distinct limitations. Faster-RCNN has a low precision (47.9%) and a large size (9.76MB), resulting in poor overall performance. EfficientNet ($\text{mAP} = 83.9\%$) and CenterNet ($R = 60.2\%$) show lower accuracy-related indicators. Although RT-DETR has a relatively high $\text{mAP} = 92.2\%$, its model size (32.4MB) is excessively large. Among the YOLO series (YOLOv11, YOLOv8, YOLOv9, YOLOv10), none can surpass PDMA-YOLO in terms of mAP, and PDMA-YOLO also has advantages in FPS and model size.

PDMA-YOLO outperforms other models, achieving an optimal balance of precision, speed, and lightweight characteristics, making it more suitable for practical applications requiring high-efficiency detection, thus enhancing its reliability when applied in teaching scenarios.

3.3 Integrating Visual and Language Models

By integrating PDMA-YOLO with a large-scale language model, PlantPal has been transformed into a multimodal educational assistant capable of processing both visual and textual information for knowledge-based question answering. The system achieves an impressive accuracy rate of 96.3%, outperforming other state-of-the-art models in the domain.

This high level of performance demonstrates PlantPal's significant advantages in educational contexts, particularly in the field

of plant pathology and pest management. By combining real-time image recognition with deep contextual reasoning, PlantPal allows students, agricultural trainees, and field practitioners to receive immediate, accurate feedback on pest and disease identification, causal analysis, and integrated pest management strategies. Moreover, the multimodal architecture supports interactive learning, enabling users to upload images of affected crops and receive explanations grounded in verified domain knowledge. This bridges the gap between theoretical learning and practical fieldwork, enhancing both engagement and knowledge retention.

As agriculture faces increasing challenges from climate change, emerging pathogens and labor shortages, intelligent systems like PlantPal provide a promising pathway for empowering the next generation of agronomists with AI-assisted tools that are both reliable and pedagogically effective.

4 Affordances

This system provides an interactive learning environment for students, enabling them to study, explore, and apply domain-specific knowledge in the fields of agriculture and forestry. The user interface supports multimodal interaction, including natural language input, image uploads, and real-time photo capture, enabling students to formulate inquiries in a conversational manner. The system responds with accurate and context-aware answers by leveraging both its built-in knowledge base and external supplementary resources.

Image-based interactions, including both uploaded images and real-time photographs, are processed using the PDMA-YOLO model developed by the research team. This model extracts semantic information related to plant diseases and pests from the images. The extracted semantic features are then combined with the user's natural language input and passed to a large language model to generate the final response.

4.1 Core Interactions

Users may query the system regarding the identification, diagnosis, or management of specific pests and diseases. Queries may range from general (e.g. "What are the common pests of tomatoes?") to highly specific (e.g. "How can whiteflies in greenhouse cucumbers be managed?"). The system interprets user intent and provides scientifically grounded and structurally organized responses.

4.2 Context-Aware Reasoning

When supplemental textual context is provided, such as field reports, case descriptions, or region-specific data, the system prioritizes this information in its reasoning process. In the absence of such input, it relies on its extensive pretrained knowledge base. In cases of conflict between the two sources, the system defers to the user-provided context to ensure situational relevance and accuracy.

4.3 Hierarchical Knowledge Navigation

Students can progressively refine their inquiries based on the system's responses. Initial query results serve as inputs for subsequent questions, enabling an exploratory learning process grounded in a step-by-step logical progression. For instance, after identifying a

suspected pest based on observed symptoms, a user can immediately inquire about its life cycle and control strategies. The system retains semantic context across interactions, thereby reducing cognitive load and reinforcing knowledge integration.

In more advanced use cases, users can construct multi-step diagnostic chains which begin with symptom-based queries, followed by pathogen identification, ecological impact assessment, and ultimately the formulation of integrated pest management (IPM) strategies. This interactive approach fosters critical thinking and encourages deeper, inquiry-driven exploration.

4.4 Reusability of Knowledge Paths

Each interaction with the system can be recorded, annotated, and revisited, enabling the creation of reusable knowledge traces. These traces can be leveraged by students or instructors for later review, peer learning, or assessment design. Furthermore, they can be extended and adapted for use in case-based reasoning exercises.

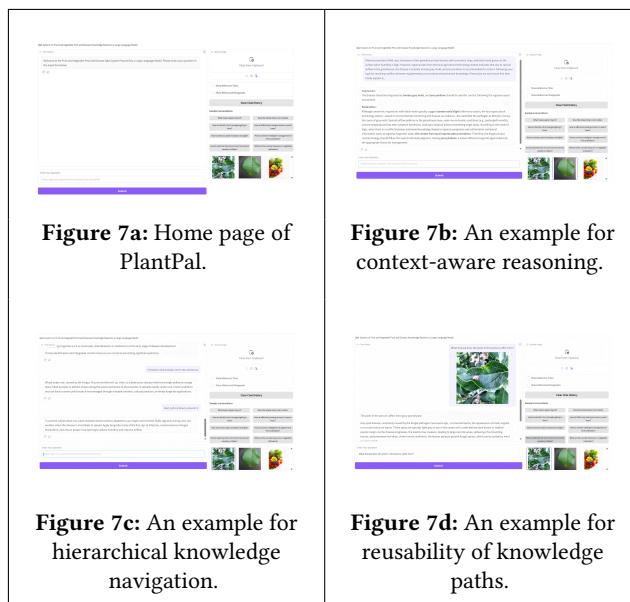


Figure 7: A compilation of several screenshots from PlantPal.

4.5 Instructional Integration

The system supports seamless integration into agricultural curricula, particularly in plant pathology and entomology courses. Instructors can design scenario-based learning activities in which students interact with the system to simulate diagnostic processes, assess pest management strategies, and reflect on the complex interactions among pests, crops, and environmental factors.

To evaluate the practical effectiveness of the system, we invited students and instructors participating in the plant pathology course to incorporate the system into weekly scenario-based learning modules. Based on post-activity surveys, a substantial majority of students indicated that the system enhanced their understanding of the complex interactions among hosts, pests, and environmental factors. Many also reported that the diagnostic simulations were

more engaging than conventional instructional methods. Instructors noted marked improvements in student participation and the depth of critical thinking demonstrated during subsequent class discussions. The user feedback underscores PlantPal's strong potential as an educational tool, particularly in fostering active learning and the application of theoretical knowledge to practical problem-solving.

5 Conclusion

In this study, we developed PlantPal, a multimodal large language model system designed to assist university students in learning about plant diseases and pests. The system integrates visual recognition through PDMA-YOLO with the reasoning capabilities of Qwen, enabling both image-based diagnosis and natural language question answering. We evaluated PlantPal on a curated dataset of plant pathology scenarios, which included both image-based cases and text-based diagnostic questions. The system achieved a high accuracy, significantly outperforming comparable models in the domain.

While existing LLMs often struggle to bridge visual recognition and domain-specific reasoning, PlantPal demonstrates strong potential for educational use, providing reliable and interpretable responses grounded in authoritative agricultural knowledge. We also observed several characteristics in PlantPal's responses, including its ability to handle ambiguous symptoms, integrate contextual sensor data, and generate actionable treatment suggestions.

Our findings suggest that multimodal LLM systems like PlantPal could play a transformative role in plant pathology education, especially in supporting interactive learning, remote diagnostics, and field-based student training. We conclude by offering practical recommendations for agricultural educators on integrating PlantPal into the curriculum, and we highlight the broader potential of such systems in promoting data-driven, accessible, and scalable agricultural education.

6 Future Work

The system has been officially approved by the National Copyright Administration of China and granted a Computer Software Copyright Registration Certificate. Building upon this, we invited a group of students from the College of Agriculture, College of Forestry and College of Plant Protection at Northwest A&F University to use the system. There has been consistently positive informal verbal feedback from students who have used the system. To substantiate these initial responses, future work will involve a systematic evaluation of the system's effectiveness and efficiency. It is anticipated that the system will be further promoted and integrated into campus-related activities. Moreover, to improve accessibility and user engagement, a dedicated website and a mobile-responsive web interface are planned for future development.

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